📞 VODACOM SWAHILI VOICE AI ASSISTANT: SYSTEM ARCHITECTURE & LOGIC FLOW

This document details the industrial-grade system architecture, database design, voice pipeline, and logical flow of the **Vodacom Swahili Customer Assistant (Ana)**. It bridges your current standard core with a fully-integrated FastAPI backend and a state-of-the-art interactive web frontend interface.

—— 1. OVERALL SYSTEM ARCHITECTURE ——

The architecture follows **Option A (Backend-Centric Neural Voice Pipeline)**. It ensures that heavy processing (Google Cloud STT speech-to-text, LLM function parsing, and ElevenLabs/Google voice synthesis) is offloaded entirely to cloud APIs, preserving 0% CPU overhead on both the client device and backend container.

```
graph TD
    %% Define Nodes
    UI[Frontend UI: Web Chat Widget]
    API[FastAPI Backend Server]
    DB[(Mock Telecom CRM Database)]
    STT[Speech-to-Text: Google Cloud STT]
    LLM[Swahili LLM Core: Groq Llama-3.3-70B]
    TTS[Text-to-Speech: ElevenLabs Victoria / Google Fallback]
    SMS[Mock SMS Gateway: Background Thread]
    HUMAN[Human Support Escalation Queue]

    %% Interactions
    UI -->|1. Post Audio WAV/WebM| API
    API -->|2. Transcribe Request| STT
    STT -->|3. Swahili Text Transcript| API
    API -->|4. Text & System Prompt| LLM

    %% Function Calling & Database Checks
    LLM -->|5. Tool Call / Action Trigger| API
    API <-->|6. Verify Profile & Ledger| DB
    API -->|7. Send Confirmation OTP| SMS

    %% Output Synthesis
    API -->|8. Swahili Response Text| TTS
    TTS -->|9. Voice Bytes Base64/MP3| API

    %% Escalation Handoff
    API -->|10. Handoff Queue with Logs| HUMAN

    %% Return to Client
    API -->|11. JSON Response: Text + Audio + ActionState| UI
```

—— 2. MOCK TELECOM DATABASE (CRM, M-PESA & NIDA) ——

To simulate a real-world telecommunication database structure, the FastAPI backend implements an in-memory SQL/JSON database containing mock data schema representing customer information, security parameters, and balance ledgers.

A. Database Schemas

1. CRM Customer Profile (`vodacom_crm`)

Field Name	Data Type	Description	Example Value
`msisdn` (PK)	`VARCHAR`	Phone number key	`"255745123456"`
`full_name`	`VARCHAR`	Account holder's legal name	`"Jane Dinah"`
`nida_number`	`VARCHAR`	National Identification Number	`"19950812-11102-00001-22"`
`sim_status`	`VARCHAR`	Current network status	`"ACTIVE"` , `"LOCKED"` , `"SUSPENDED"`
`puk_code`	`VARCHAR`	PUK code for manual unlock	`"87654321"`

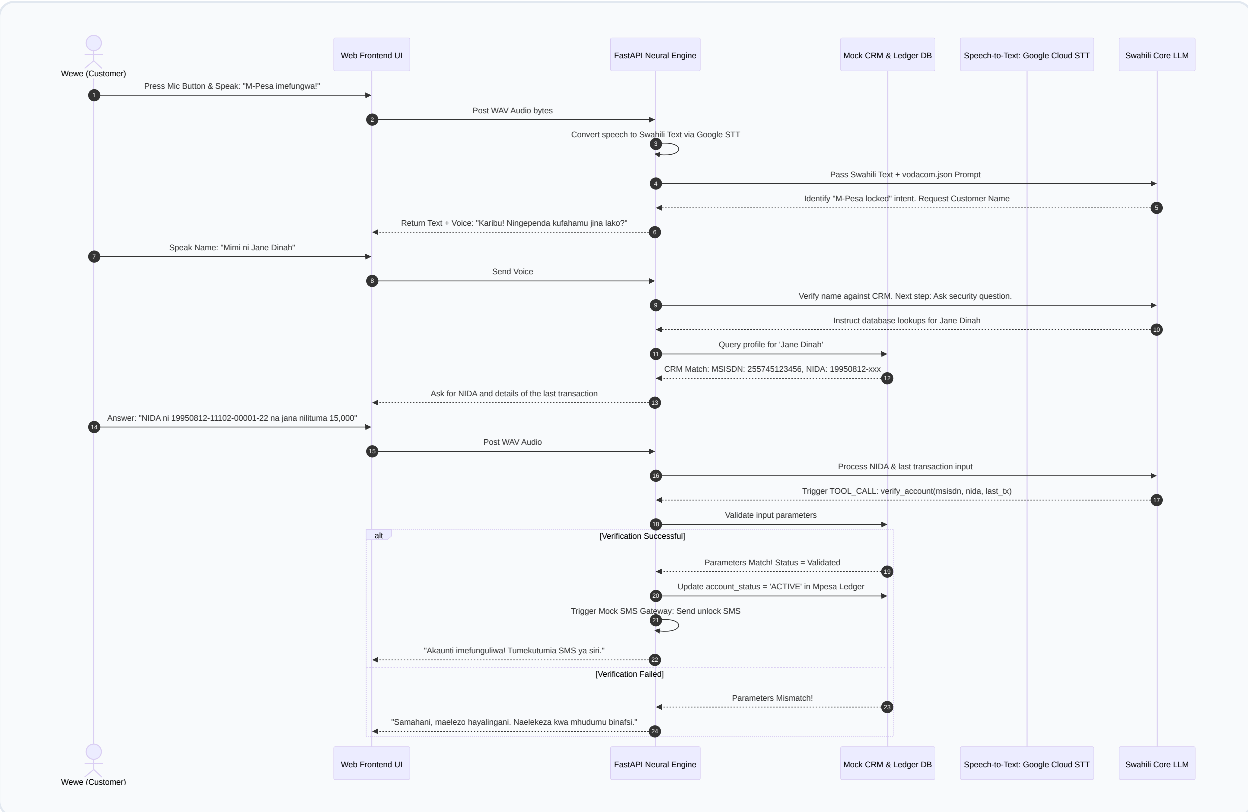
2. M-Pesa Account Ledger (`mpesa_ledger`)

Field Name	Data Type	Description	Example Value
`msisdn` (FK)	`VARCHAR`	Reference key to CRM profile	`"255745123456"`
`account_status`	`VARCHAR`	Current M-Pesa lock state	`"ACTIVE"` , `"BLOCKED"` , `"PIN_LOCKED"`
`balance_tsh`	`DECIMAL`	Available account balance	`85000.00`
`last_transaction`	`VARCHAR`	Verification parameter (Last action)	`"Sent 15,000 Tsh to John"`

— 3. VOICE CONVERSATIONAL FLOW & SECURITY LOGIC —

Telecom customer assistance operates on a rigid **Security & Empathy Verification Protocol**. Before resolving account locks or executing transaction reversals, the AI must establish trust and verify identity, just like a live Vodacom customer support agent.

🔄 The Customer Service Verification Loop



— 4. FRONTEND UI SPECIFICATION & MICRO-ANIMATIONS —

To ensure the client-side experience feels highly premium and premium-grade, the UI layout incorporates curated design tokens representing **Vodacom's signature Red branding** wrapped in a sleek glassmorphic container.

🎨 Design Tokens & UI Elements

- **Harmonious Color Palette:**
  - `primary-red` : `hsl(0, 100%, 45%)` (Vibrant Vodacom red accent)
  - `dark-bg` : `hsl(220, 15%, 8%)` (Deep midnight slate theme)
  - `glass-surface` : `hsla(220, 15%, 15%, 0.6)` (Translucent container overlay)
  - `border-glow` : `hsla(0, 100%, 45%, 0.35)` (Subtle red perimeter halo)

- **Typography:** Google Font **Outfit** (`font-family: 'Outfit', sans-serif`) for crisp, modern legibility.

🎙️ The Floating Vocal UI Component

The interface features a floating circular microphone button surrounded by multi-layered SVG rings that respond with dynamic micro-animations to user interaction.

```css
/* Glassmorphic Chat Widget Container */
.ana-chat-container {
    background: rgba(18, 20, 24, 0.75);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border: 1px solid rgba(230, 0, 0, 0.2);
    border-radius: 24px;
    box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.5), 0 0 15px rgba(230, 0, 0, 0.1);
}

/* Premium Vocal Recording Button */
.mic-trigger-btn {
    width: 72px;
    height: 72px;
    border-radius: 50%;
    background: radial-gradient(circle, #e60000 0%, #990000 100%);
    border: none;
    cursor: pointer;
    position: relative;
    transition: all 0.4s cubic-bezier(0.175, 0.885, 0.32, 1.275);
}

/* Pulsing Outer Waves Visualizer (Animated during Recording) */
.mic-trigger-btn.recording::after {
    content: '';
    position: absolute;
    top: -12px;
    left: -12px;
    right: -12px;
    bottom: -12px;
    border-radius: 50%;
    border: 2px solid #e60000;
    opacity: 1;
    animation: wave-ping 1.5s cubic-bezier(0, 0, 0.2, 1) infinite;
}

@keyframes wave-ping {
    75%, 100% {
        transform: scale(1.4);
        opacity: 0;
```

```
    }
}
```

---

5. HUMAN ESCALATION & HANDOFF PROTOCOL

When a complex scenario emerges (e.g., three consecutive verification failures, customer requests a supervisor, or fraud-alert flag triggers), the system switches seamlessly to **Human Escalation State**.

Handoff Logic Pipeline:

1. **Freeze AI Responses:** The backend pauses LLM completions for this MSISDN.
2. **Generate Transcript Brief:** A background worker compiles the conversation history, security status, and verification log into a structured JSON package.
3. **Active UI State Change:** The Frontend UI changes color (from Vodacom Red to Slate Blue), displaying: > *"Mawasiliano yameelekezwa kwa Mhudumu Binafsi (Human Agent). Tafadhali subiri kidogo..."*
4. **Handoff API Notification:** The backend publishes the session state to an internal WebSocket queue for live customer service agents.

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6. BACKGROUND MOCK SMS GATEWAY PIPELINE

For premium, real-world customer assurance, transactions and unlocks must trigger confirmation notifications. The FastAPI backend employs a **Python Asyncio Event Loop Queue** to simulate background carrier interactions.

```
import asyncio

async def trigger_mock_sms_gateway(msisdn: str, message: str):
    """
    Simulates sending an active SMS transaction alert/OTP to a customer's phone
    via carrier API without stalling the core chat UI response time.
    """
    print(f"[SMS Gateway] Kuanzisha utumaji wa ujumbe kwenda: {msisdn}")
    # Simulate carrier latency (1.5 seconds)
    await asyncio.sleep(1.5)
    print(f"[SMS Gateway SUCCESS] Ujumbe Umetumwa! Content: '{message}'")
```

—— 7. ELEVENLABS VOICE SYNTHESIS & GOOGLE FALLBACK ENGINE ——

The text-to-speech engine ensures that the voice output is generated in Victoria's highly realistic tone. The backend utilizes pure standard library constructs to maximize request speed.

```
import urllib.request
import json
import os

def synthesize_swahili_voice(text: str, provider: str = "elevenlabs") -> bytes:
    """
    Processes Swahili text and synthesizes speech.
    If ElevenLabs API key limit (HTTP Error 402) occurs, it automatically falls b
    to Google Cloud Neural TTS to guarantee uninterrupted voice availability.
    """
    if provider == "elevenlabs":
        api_key = os.environ.get("ELEVENLABS_API_KEY")
        voice_id = os.environ.get("ELEVENLABS_VOICE_ID", "FZGeNF7bE3syeQOynDKC")
        url = f"https://api.elevenlabs.io/v1/text-to-speech/{voice_id}"

        payload = {
            "text": text,
            "model_id": "eleven_multilingual_v2",
            "voice_settings": {"stability": 0.5, "similarity_boost": 0.75}
        }

        req = urllib.request.Request(
            url,
            data=json.dumps(payload).encode("utf-8"),
            headers={"xi-api-key": api_key, "Content-Type": "application/json"},
            method="POST"
        )
        try:
            with urllib.request.urlopen(req) as response:
                return response.read() # Return raw MP3 audio bytes
        except Exception as e:
            print(f"[!] ElevenLabs Synthesis failed: {e}. Switching to Google TTS
            # Fallback to Google
            return synthesize_swahili_voice(text, provider="google")

    # Google Cloud Translate TTS fallback path
    # (Extracts, merges, and streams audio from public Google TTS)
    # ...
```

> *[!TIP] **Implementation Recommendation:** When building your final FastAPI backend routes, initialize standard CORS middleware to enable secure browser-based microphone uploads from your HTML frontend client. Use `multipart/form-data` to submit audio blobs directly.*